

Raghav Nandwani

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Visa : Currently eligible to relocate and work in Germany through the Chancenkarte (Opportunity Card).

PROFILE SUMMARY

Engineering Manager and Applied AI practitioner with 5+ years of experience leading the development, optimization, and deployment of AI systems across speech, language, vision, and multimodal domains. Experienced in translating cutting-edge research into scalable production solutions by driving model benchmarking, training, fine-tuning, MLOps, and AI platform development. Proven track record of delivering end-to-end AI products, including conversational AI, speech technologies, personalization systems, and edge AI deployments, while collaborating across research, engineering, and product organizations. Passionate about building robust AI platforms that enable rapid experimentation, reliable deployment, and measurable business impact.

CORE TECHNICAL EXPERTISE

- **Applied AI Research & Experimentation:** Model prototyping, benchmarking, ablation studies, performance evaluation on real-world datasets
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, transformer-based architectures, multimodal perception and learning
- **Robotics & Physical AI:** Autonomous navigation, localization, dense mapping, policy learning/execution
- **Edge AI & Embedded Systems:** NVIDIA Jetson AGX Orin, Isaac ROS, TensorRT, CUDA, real-time inference systems
- **ML Systems & Optimization:** Quantization, pruning, accelerator-aware tuning, inference profiling, deployment pipelines
- **Software & Infrastructure:** Python, C++, ROS 2, Docker, Kubernetes, Databricks

PROFESSIONAL EXPERIENCE

Jio Platforms
Tech Lead (Core AI)

Oct 2024 – Present
Mumbai, India

- **Lead the Core AI group** responsible for **benchmarking, training, optimizing, and deploying** state-of-the-art AI models powering Jio's device ecosystem, including AI platforms, smart glasses, robotics, and multimodal user experiences.
- Drive the **end-to-end development** and productionization of AI capabilities across speech, language, and vision domains, collaborating with research, platform, and product teams to deliver scalable AI solutions.
- Evaluated, fine-tuned, and deployed a diverse portfolio of AI models, including Wake Word Detection, Language Identification (LID), Speech-to-Text (STT), Voice Cloning and Text-to-Speech (TTS), and Visual Question Answering (VQA) systems.
- **Improved Language Identification (LID) accuracy** for Indian languages from ~78% benchmark performance to 92% using internally developed models, delivering a 14-percentage-point improvement over leading cloud-based alternatives.
- **Architected and developed a low-latency Voice-to-Voice agentic AI platform** integrating Automatic Speech Recognition (ASR), LID, Speaker Diarization, Large Language Models (LLMs), and TTS pipelines, currently serving over 10,000 requests per hour with architecture designed to scale beyond 1 million requests per day.
- Established **MLOps workflows** and automated model delivery pipelines, reducing deployment turnaround time by 75% and enabling rapid, reliable delivery of AI models across multiple product teams.
- Designed and implemented a **Databricks-based data platform** following **Medallion Architecture** principles, enabling scalable data processing, experimentation, analytics, and AI lifecycle management.
- Developed **observability frameworks** and analytics dashboards providing technical and product metrics to monitor AI platform performance, reliability, adoption, and user engagement.
- Built **platform-level AI capabilities** including semantic query caching, AI-powered note-taking, memory systems (short-term and long-term memory), and user personalization frameworks to improve contextual and adaptive AI interactions.
- Led benchmarking and optimization efforts for AI inference workloads across **cloud and edge environments**, balancing latency, accuracy, and resource utilization for production deployments.
- Explored and evaluated emerging **multimodal and embodied AI models** for next-generation intelligent devices, assessing their feasibility for production deployment across Jio's AI ecosystem.

- Conducted **applied research in on-device perception and scene understanding** for XR platforms under real-time and power constraints.
- Improved detection and segmentation accuracy by ~25% through **architectural experimentation, transformer-based feature fusion, and multimodal learning approaches**.
- Benchmarked model performance across latency, accuracy, and hardware efficiency to guide **design decisions for deployment on edge accelerators**.
- Collaborated with hardware and systems teams to **translate research models into optimized, production-ready implementations**.
- Mentored engineers on **applied ML research workflows, experimentation methodologies, and system-level optimization**.

- Prototyped and evaluated **real-time perception and localization systems** for autonomous agricultural vehicles in unstructured environments.
- Conducted **field validation and robustness testing** of perception pipelines under diverse environmental conditions.
- Optimized models for edge deployment through **iterative experimentation on accuracy–latency tradeoffs**.
- Collaborated with autonomy engineers to **validate system performance against real-world vehicle behavior**.

- Developed and deployed real-time computer vision systems in customer-facing production environments, emphasizing low-latency inference, robustness, and operational reliability.
- Partnered with product and platform teams to translate customer requirements into scalable, production-ready perception pipelines.

- Deployed edge-optimized computer vision models for production applications, balancing accuracy, latency, and resource constraints.
- Improved inference performance through profiling and architecture-level optimization for efficient deployment.

EDUCATION
Master of Engineering (M.Eng.) — Robotics & Artificial Intelligence
University of Maryland, College Park | 2018 – 2020

Bachelor of Technology (B.Tech.) — Mechatronics Engineering
Manipal University Jaipur | 2013 – 2017

CERTIFICATIONS

- AWS Technical Essentials — Amazon Web Services
- Machine Learning Pipelines on Cloud — Amazon Web Services